Jaiveer Bassi

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Experience

Stealth Robotics Startup (Y Combinator) Oct 2026 – Present
Software Engineer (Part-Time) *Menlo Park, CA*

- Built the browser-based operator console used to run teleoperation and policy evaluations on bimanual robot stations, including live camera feeds, episode control, scoring, and authentication.
- Reduced camera feed latency from 200 ms to as low as 20 ms and resolved a socket failure in the station control daemon.
- Brought up the console on additional robot stations by diagnosing hardware and configuration faults.
- Designed a six-DOF Dynamixel GELLO-style leader arm in CAD and delivered printable parts, firmware, and an assembly guide.

Physical Intelligence May 2026 – Present
AI Robot Operator II *San Francisco, CA*

- Triage recurring hardware-software faults across the robotics stack, trace service logs to isolate root causes, and document reproducible findings for engineering teams.
- Evaluate policy rollouts against task criteria, identify failure modes and edge cases, annotate executions, and enforce data-quality thresholds used in model development.
- Generate high-quality demonstration data for general-purpose robotics foundation models by teleoperating robotic arms through manipulation, sorting, and assembly tasks.

Stanford University Sep 2026 – Present
Stanford SSG Fellow *Stanford, CA*

- Conduct mentor-guided independent research by developing a core curiosity question, evaluating primary scientific literature, and formulating testable hypotheses.

Handshake AI Apr 2026 – Jun 2026
Contracted AI Response Evaluator *Remote*

- Evaluated LLM-generated code and technical responses across Python, C#/.NET, AI/ML, and software engineering for correctness, instruction following, and technical clarity.

Castline Studio Jan 2026 – Apr 2026
Operator & Developer – Resin 3D Printing, Next.js, TypeScript *Vallejo, CA*

- Operated resin 3D-printing production for an e-commerce business generating $2K+/month across 100+ product listings, including calibration, curing, post-processing, and tolerance control.
- Built an order-management platform with client-side STL parsing for automated geometry, volume, dimension, and print-cost calculations.

Kaiser Permanente Regional Laboratory Oct 2023 – Jun 2024
Software Developer *Berkeley, CA*

- Developed a full-stack application using React, Express, RESTful APIs, and SQL for uploading, processing, and reviewing clinical laboratory instrument data across multiple departments.
- Added validation and error logging to a 50+ column instrument-data workflow; diagnosed faulty recordings and documented recurring failure patterns to improve pipeline reliability.

Selected Projects

CAN Bus Fault-Injection & Diagnostics Rig | *C++, Arduino, MCP2515 (SPI), SocketCAN*

- Assembled a three-node CAN network of Arduino microcontrollers driving MCP2515 CAN controllers over SPI, with a USB-CAN interface and a custom wiring harness, then induced bus-off, arbitration, connector, wiring, and termination faults.
- Mapped physical failure modes to reproducible SocketCAN error states and multimeter evidence in a published diagnostic catalog.

Scribbot Desktop Drawing Arm | *C++, Arduino Nano, PCA9685 (I2C), OpenSCAD*

- Designed in OpenSCAD and built a 3D-printed four-servo drawing arm controlled by an Arduino Nano driving a PCA9685 PWM controller over I2C.

GELLO-Style Leader-Follower Teleoperation Rig | *Arduino, ADC Sensing, Servo Control*

- Built a four-DOF leader-follower mechanism for approximately $30 by assembling linkages, wiring potentiometer-based joint encoders, and calibrating direct joint-space control without inverse kinematics.

π_0 **on a Budget** | *OpenPI, LoRA, LeRobot, JAX, Arduino*

- Built a synchronized data pipeline recording 30 FPS video and joint commands from a custom 4-DOF teleoperation arm, with conversion into a LeRobot-compatible dataset.
- Implemented an RTX 4070 workflow for π_0-FAST fine-tuning, memory profiling, policy serving, closed-loop evaluation, and a real-time dashboard of live Arduino telemetry and predicted action chunks.

UR5e Fault Injection & Diagnostics Harness | *RTDE, URSim, Fault Injection*

- Built a control and fault-characterization harness that drives simulated robot motion and logs synchronized joint telemetry while testing connection failures, stale handles, hangs, and native crashes.

Research Publications

Smoothness Ranks Skill, Not Success: Auditing Trajectory-Smoothness Curation Metrics

- Showed that published smoothness metrics for curating robomimic demonstrations mostly track episode length and operator skill; curating by TED cut downstream policy success by a third versus random selection. Accepted (poster) at the NeurIPS 2026 RoboPAD workshop; arXiv:2610.17427.

Not All Bad Demonstrations Are Equally Bad

- Ran 660 controlled policy fits across five demonstration failure modes, two policy families, ten seeds, and transition-matched controls; under review at the CoRL 2026 LfC workshop.

What Can Three Seeds and Fifty Rollouts Resolve? Operator and Seed Variance on robomimic

- Preregistered audit of 138 BC-RNN runs; at 50 rollouts ten seeds showed no detectable difference on any task, while 500 rollouts exposed the seed effect and erased an apparent operator-proficiency pattern; under review at the CoRL 2026 SimBench2Real workshop.

One Byte, One Rank Reversal: Auditing HumanEval Pipeline Sensitivity

- In a pinned five-model RTX 4070 reproduction, StarCoder2-3B scored 3/164 against its published 31.7%; removing one trailing prompt newline raised it to 49/164 and restored the ranking.

When Does INT8 Actually Accelerate Host-CPU Inference?

- Audited MobileNetV2 INT8 conversion and packaging, showing that INT8 latency depends on backend delegation and quantization granularity; v1.2.0 archived on Zenodo.

Service

Reviewer, CoRL 2026 Workshop on Learning from Corrections and Interventions (LfC) 2026

Technical Skills

Languages: C++, Python, C#, TypeScript, JavaScript, SQL, Bash
Embedded & Hardware: Arduino, microcontrollers, SPI, I2C, CAN/SocketCAN, PWM servo and actuator control, sensor wiring, PCB rework, soldering, multimeter diagnostics
Robotics & Systems: RTDE, URSim, ROS2, gRPC, real-time control, fault injection, telemetry analysis, Linux, Git, Docker, Kubernetes, PostgreSQL
Machine Learning: PyTorch, JAX, LeRobot, OpenPI, LoRA, ONNX, NumPy, model evaluation, CUDA

Education

Grand Canyon University May 2024 – Oct 2025
M.S. in Computer Software Engineering *Phoenix, AZ*

- Capstone: Developed an AI model-packaging CLI that converts PyTorch, TensorFlow, ONNX, and scikit-learn models into deployable Docker images or Python packages.

University of Silicon Valley May 2021 – Aug 2023
B.S. in Computer Science *San Jose, CA*

- Capstone: Developed a MetaHuman lip-sync and audio-animation pipeline in Unreal Engine for real-time playback.

Certifications

Accelerated Computing in Modern CUDA C++ (Sep 2026)
Cisco Network Support and Security (Sep 2025)
Stanford AI and Machine Learning in Healthcare (Jun 2025)
HackerRank Software Engineer Certificate (Mar 2026)
IBM Artificial Intelligence Fundamentals (Aug 2025)
Google IT Support Professional Certificate (May 2021)